

# LARGE ERDŐS–STRAUS WITNESSES: Ω-RESULTS FOR THE MULTIPLIER AND SLICE PARAMETERS

ANONYMOUS

**ABSTRACT.** For a prime  $p$  let  $W(p)$  be the least modulus  $M \equiv 3 \pmod{4}$  of a “multiplier” congruence that forces a solution of  $4/p = 1/x + 1/y + 1/z$ , and let  $ck_{\min}(p)$  be the least product  $ck$  of a positive Type I slice. Both are expected to be at most a power of  $\log p$  for almost all  $p$ , and are known to exceed a constant times  $\log p$  for infinitely many  $p$ . We prove that  $W(p) \geq (\log p)^2 \exp(-C \log \log p / \log \log \log p)$  for infinitely many primes  $p$  in Mordell’s hard classes. The construction is a prime-local sieve: the class of one is imposed modulo all small primes, a congruence saving bounds the residual sieve mass by  $T^{o(1)}$ , and the sieve is transferred to primes by the Linnik-range prime number theorem of Thorner and Zaman, whose relative error term absorbs a possible Siegel zero. We show that exponent 2 is the limit of every prime-local design and isolate a purely combinatorial hypothesis that suffices to go further. For the slice parameter we prove that congruence methods certify exactly the least quadratic non-residue, giving  $ck_{\min}(p) \gg \log p \log \log \log p$  infinitely often (via Lau–Wu, after Graham–Ringrose) and showing that a power saving would be a breakthrough on the least non-residue. On the Haar side we prove that the avoider density satisfies  $\log(1/\delta^*(T)) \leq T^{1/3+o(1)}$ .

## 1. INTRODUCTION

The Erdős–Straus conjecture asserts that  $4/n = 1/x + 1/y + 1/z$  is solvable in positive integers for every  $n \geq 2$ . It suffices to treat primes, it has been verified for  $n \leq 10^{18}$  ([10]; [13] for  $10^{17}$ ; see [12], §6), and Mordell’s identities settle every prime outside the six classes  $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$  [9] (cf. also [11]); we call such primes *hard*. On the exceptional-set side, Vaughan [15] proved that the number  $E(N)$  of  $n \leq N$  without a representation satisfies  $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$ ; see [12] for a uniform modern account and for many exceptions to the analogous statement with large numerators, and [4] for the average number of solutions, which over primes  $p \leq N$  is  $(\log N)^3$  up to a factor  $\log \log N$ .

This note is about the *pointwise size* of the smallest witness. Two natural witness parameters are the following.

*The multiplier parameter.* For  $M \equiv 3 \pmod{4}$  put  $A_M = (M + 1)/4$ . If  $uvw = A_M$  and  $n \equiv -uv^{-1} \pmod{M}$ , then  $s = (nv + u)/M$  is a positive

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integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw} \quad (1)$$

(direct verification; this is the classical multiplier family, cf. [15, 4]). Define

$$W(n) = \min\{M \equiv 3 \pmod{4} : n \equiv -uv^{-1} \pmod{M} \text{ for some } uvw = A_M\}, \quad (2)$$

with  $W(n) = +\infty$  if no such  $M$  exists. Thus  $W(p) \leq T$  supplies an explicit representation with a modulus at most  $T$ .

*The slice parameter.* A Type I solution with parameters  $(a, b, c, k)$  satisfies  $p(a+b) = k(4abc-1)$  and gives  $4/p = 1/(ack) + 1/(bck) + 1/(pabc)$ . For an admissible pair  $(c, k)$  (see §8) let  $M_{c,k}(p)$  be the number of divisors of  $p^2 + 4ck^2$  lying in the class  $-p \pmod{4ck}$ ; the slice is positive iff  $M_{c,k}(p) > 0$  (see §8). Put  $ck_{\min}(p) = \min\{ck : M_{c,k}(p) > 0, \text{sf}(c) \notin \{1, 2, 3, 6\}\}$  (the cores 1, 2, 3, 6 are excluded because they are deterministic on hard primes).

Both parameters are expected to be small for almost all primes. For  $W$ , tails of the form  $\#\{p \leq N : W(p) > T\} \ll N \exp\{-c(\log T)^2 \log \log T\}$  (in the range  $\log N \gg (\log T)^3 \log \log T$ ) are obtained in companion work whose assembly is still under internal review; we do not use them. In the other direction, the class of one (every prime  $p \equiv 1 \pmod{\text{lcm}\{M \leq T\}}$ ) has  $W(p) > T$  together with Linnik's theorem shows  $W(p) \gg \log p$  infinitely often. With the smaller modulus  $\text{lcm}(24, M \leq T, M \equiv 3 \pmod{4})$  and Chang's least-prime bound for smooth moduli [2], the constant becomes  $5/8$  (§7). The same is true for  $ck_{\min}$  via genus characters. Whether  $W(p)/\log p$  is unbounded was open.

**Theorem 1.1** (Proved modulo Theorem 4.1; effective). *There is an absolute constant  $C$  such that for infinitely many hard primes  $p$*

$$W(p) \geq (\log p)^2 \exp\left(-C \frac{\log \log p}{\log \log \log p}\right).$$

*More precisely, for every sufficiently large  $T$  there is a prime  $p \equiv 1 \pmod{840}$  with  $W(p) > T$  and  $\log p \leq T^{1/2} \exp(O(\log T / \log \log T))$ .*

In particular the statement “ $W(p) \leq (\log p)^A$  for all large  $p$ ” is false for every  $A < 2$ . The cited input (Theorem 4.1) is Thorner and Zaman's prime number theorem in the Linnik range [14], including the McCurley zero-free region statement quoted there. No hypothesis on Siegel zeros is needed: the exceptional zero is absorbed by a uniqueness lemma across the family of moduli (Lemma 4.2) and a two-case analysis (Theorem 5.1).

The exponent 2 is the natural limit of the method (Proposition 7.3). Beyond it, this route needs a pointwise minorant that handles atoms with several large prime factors (a “hypergraph” minorant); Theorem 7.5 shows that such a minorant suffices, the prime side being completely handled.

For the slice parameter the picture is different: the only congruence mechanism is the genus character, and it certifies exactly the least quadratic non-residue  $n_p$ .

**Theorem 1.2.** (a) (Proved.)  $ck_{\min}(p) \geq n_p$  for every prime  $p \equiv 1 \pmod{24}$ , and every fixed-period congruence minorant for  $\{ck_{\min}(p) > T\}$  is supported, among large primes, on primes with  $n_p > T$  (Corollary 8.5).  
 (b) (Proved modulo [7, Prop. 5.1].)  $ck_{\min}(p) \gg \log p \cdot \log \log \log p$  for infinitely many hard primes.

Thus a congruence proof of  $ck_{\min}(p) > (\log p)^{1+\delta}$  infinitely often would improve the best known unconditional (Graham–Ringrose) and GRH-conditional (Montgomery)  $\Omega$ -results for  $n_p$ , and would exceed the random-model prediction  $\max_{p \leq x} n_p \asymp \log x \log \log x$ .

Finally, on the Haar (profinite) side, let  $\delta^*(T)$  be the Haar measure of the set of  $n \in \widehat{\mathbb{Z}}^\times$  with  $n \equiv 1 \pmod{24}$  and  $W(n) > T$ , normalised within the class  $1 \pmod{24}$ . The class of one gives  $\log(1/\delta^*(T)) \leq (2/3 + o(1))T$ , and a crude prime-local bound gives  $T^{1/2+o(1)}$ .

**Theorem 1.3** (Proved).  $\log(1/\delta^*(T)) \leq T^{1/3+o(1)}$ .

We do not pursue heuristics about the true size of  $\delta^*(T)$  here.

**Status conventions.** Every theorem header carries a label. *Proved* means proved in full here. *Proved modulo X* means proved here assuming the cited statement X, whose statement we checked in the source but whose proof we did not check. *Cited* marks external statements.

## 2. THE ATOMS OF THE MULTIPLIER PROBLEM

For  $M \equiv 3 \pmod{4}$  put  $\mathcal{R}(M) = \{-4D \bmod M : D \mid A_M^2\}$ .

**Lemma 2.1** (Proved).  $\{-uv^{-1} \bmod M : uvw = A_M\} = \mathcal{R}(M)$ . Consequently, for every integer  $n$ ,

$$W(n) > T \iff n \bmod M \notin \mathcal{R}(M) \text{ for every } M \leq T, M \equiv 3 \pmod{4}.$$

*Proof.* Since  $4A_M \equiv 1 \pmod{M}$  and  $\gcd(A_M, M) = 1$ , from  $uvw = A_M$  we get  $v^{-1} \equiv 4uw$ , hence  $-uv^{-1} \equiv -4u^2w$  and  $D = u^2w$  divides  $A_M^2$ . Conversely let  $D \mid A_M^2$ . For each prime  $q$  with  $q^a \parallel A_M$  and  $q^d \parallel D$  ( $d \leq 2a$ ) put  $u_q = \max(0, d-a)$ ,  $w_q = d-2u_q$ ,  $v_q = a-u_q-w_q$ ; these are non-negative, so  $u = \prod q^{u_q}$ ,  $v = \prod q^{v_q}$ ,  $w = \prod q^{w_q}$  satisfy  $uvw = A_M$  and  $u^2w = D$ .  $\square$

**Lemma 2.2** (Class of one; proved).  $1 \notin \mathcal{R}(M)$  for every  $M \equiv 3 \pmod{4}$ .

*Proof.* If  $-4D \equiv 1$  then  $D \equiv -A_M \pmod{M}$ . The map  $D \mapsto A_M^2/D$  preserves this congruence, so we may assume  $D \leq A_M$ ; then  $0 < D + A_M \leq 2A_M < M$  while  $M \mid D + A_M$ .  $\square$

## 3. THE PRIME-LOCAL RESIDUAL SYSTEM

Fix  $T$  large and put  $\mathcal{L} = \log T$ ,

$$y = \sqrt{T} \exp(3\mathcal{L}/\log \mathcal{L}),$$

$$Q = \text{lcm}(24, \ell^{e_\ell} : \ell \leq y \text{ prime}, \ell^{e_\ell} \leq T \text{ maximal}),$$

and let  $\mathcal{U}$  be the set of primes in  $(y, T]$ . Then  $\log Q \leq 2y$  for large  $T$  (only  $\ell \leq \sqrt{T}$  have  $e_\ell \geq 2$ ), and  $840 \mid Q$ . For  $\ell \in \mathcal{U}$  let

$$F_\ell = \{-4D \bmod \ell : m \leq T/\ell, m\ell \equiv 3 \pmod{4}, D \mid A_{m\ell}^2, m \mid 4D+1\},$$

$$g(\ell) = \frac{|F_\ell|}{\ell-1}, \quad S = \sum_{\ell \in \mathcal{U}} g(\ell).$$

All elements of  $F_\ell$  are units.

**Lemma 3.1** (Prime-local reduction; proved). *If  $n \equiv 1 \pmod{Q}$  and  $n \bmod \ell \notin F_\ell$  for all  $\ell \in \mathcal{U}$ , then  $W(n) > T$ .*

*Proof.* Let  $M \leq T$ ,  $M \equiv 3 \pmod{4}$ . If every prime factor of  $M$  is at most  $y$ , then  $M \mid Q$ , so  $n \equiv 1 \pmod{M}$  and Lemma 2.2 applies. Otherwise let  $\ell > y$  divide  $M$ ; since  $\ell^2 > T$  we have  $\ell \parallel M$ , and  $m = M/\ell < T/y \leq y$ , so  $m \mid Q$  and  $n \equiv 1 \pmod{m}$ . If  $n \equiv -4D \pmod{M}$  with  $D \mid A_M^2$ , then  $m \mid 4D+1$  and  $n \bmod \ell \in F_\ell$ . Now use Lemma 2.1.  $\square$

**Lemma 3.2** (Local sizes; proved).  *$g(\ell) \leq 2T\tau^*(T)^2/y^2$ , where  $\tau^*(T) = \max_{n \leq T} \tau(n)$ . In particular  $\max_\ell g(\ell) \leq 1/16$  for large  $T$ .*

*Proof.* For each  $m \leq T/\ell$  there are at most  $\tau(A^2) \leq \tau(A)^2$  choices of  $D$ , and  $1/(\ell-1) \leq 2/\ell$ . Since  $\log \tau^*(T) \leq (\log 2 + o(1))\mathcal{L}/\log \mathcal{L}$  [6, Thm. 317], the bound is  $\leq 2 \exp((2 \log 2 - 6 + o(1))\mathcal{L}/\log \mathcal{L})$ .  $\square$

The next lemma is the key arithmetic input: the congruence  $m \mid 4D+1$  saves a factor  $m$ .

**Lemma 3.3** (Mass bound; proved). *With  $X = (T+1)/4$ ,*

$$S \leq \frac{4}{3}(3 + \log X) \sum_{\substack{sr^2 \leq X \\ s \text{ squarefree}}} \frac{\tau(4sr^2 + 1)}{sr}$$

$$\leq 2(3 + \mathcal{L})(1 + \mathcal{L})^2 \tau^*(T+2) = \exp\left(O\left(\frac{\log T}{\log \log T}\right)\right).$$

*Proof.* Clearly  $S \leq \sum 1/(\ell-1)$  over triples  $(m, \ell, D)$  as in the definition of  $F_\ell$ ; put  $A = A_{m\ell}$ . Since  $4A \equiv 1$  and  $4D \equiv -1 \pmod{m}$ , we have  $4A^2D^{-1} \equiv -1 \pmod{m}$ , so  $D \mapsto A^2/D$  preserves  $m \mid 4D+1$  and exchanges  $D > A$  with  $D < A$ ; hence it suffices to count  $D \leq A$  and double. Write  $D = sr^2$  with  $s$  squarefree. Then  $D \mid A^2$  iff  $sr \mid A$ ; write  $A = srk$ , so that  $D \leq A$  iff  $k \geq r$ . The map  $(m, \ell, D) \mapsto (m, s, r, k)$  is injective, and  $1/(\ell-1) \leq 2/\ell = 2m/(4A-1) \leq 2m/(3A)$ .

From  $m \mid 4sr^2 + 1$  and  $m \mid m\ell = 4srk - 1$  we get  $m \mid 4sr(r+k)$ , and  $\gcd(m, 4sr) = 1$ , so

$$m \mid r+k.$$

Since  $k \geq r \geq 1$ , the least admissible  $k$  is at least  $\max(r, m-r) \geq m/2$ , so  $\sum_{k \equiv -r(m), k \leq X} 1/k \leq (3 + \log X)/m$ . Summing over  $m \mid 4sr^2 + 1$  gives the first bound; the second follows from  $\sum_{s \leq X} 1/s \cdot \sum_{r \leq \sqrt{X}} 1/r \leq (1 + \mathcal{L})^2$  and  $4sr^2 + 1 \leq T+2$ .  $\square$

**Remark 3.4.** Without the congruence  $m \mid r + k$  the same count gives  $S \leq T^{1/2-\eta+o(1)}$  for  $y = T^{1/2+\eta}$ , and the transfer below then only yields  $W(p) \gg \log p$ . Numerically  $S \approx 6.5, 13.6, 23.9, 38.5$  for  $T = 10^3, \dots, 10^6$  (with  $y = \sqrt{T}$ ), i.e.  $S$  is polylogarithmic in practice. These values illustrate  $S$  only: the parameter  $y = \sqrt{T}$  does *not* instantiate Theorem 1.1. At  $y = \sqrt{T}$  one has  $\max g \approx 0.08\text{--}0.33 > 1/16$ , and the twist condition of §6 can fail (numerically it does at  $T = 10^4$ ). The bound  $g \leq 1/16$  of Lemma 3.2, and hence Theorem 1.1, is asymptotic: with the  $y$  above,  $y < T$  (so that  $\mathcal{U} \neq \emptyset$ ) only when  $\log T > e^6 \approx 403$ .

#### 4. THE ANALYTIC INPUT

**Theorem 4.1** (Thorner–Zaman [14, Cor. 1.4 and Rem. 1.5]; cited). *There are absolute effectively computable constants  $c_4 > 0$ ,  $C_0$  such that for coprime  $a, q$  with  $q \geq 2$  and  $x \geq q^{12}$ ,*

$$\sum_{\substack{p \leq x \\ p \equiv a \pmod{q}}} \log p = \frac{\lambda x}{\varphi(q)} (1 + \varepsilon),$$

$$|\varepsilon| \leq C_0 \left( e^{-c_4 \log x / \log q} + e^{-c_4 (\log x)^{3/5} / (\log \log x)^{1/5}} \right),$$

where  $\lambda = 1 - \chi_1(a)x^{\beta_1-1}/\beta_1$  if the exceptional zero  $\beta_1$  for the modulus  $q$  exists and  $\lambda = 1$  otherwise. Here, following McCurley [8] as quoted in [14, p. 1],  $\prod_{\chi \bmod q} L(s, \chi)$  has no zero in  $\Re s \geq 1 - 1/(13 \log(q(|\Im s| + 3)))$  apart from at most one real zero  $\beta_1$ , which is simple and belongs to a unique real character  $\chi_1$ ; also  $0 < \lambda < 2$ .

We used the arXiv version; we did not compare the numbering with the published version. We also use the quoted McCurley statement for moduli other than those of Theorem 4.1; alternatively the classical Landau–Page theorem [3, Ch. 14] with a constant  $c \leq 1/13$  in place of  $1/13$  works verbatim.

**Lemma 4.2** (One severe character per family; proved modulo the McCurley statement). *Let  $\mathcal{Q}$  be a finite set of moduli, all  $\leq Z$ . Call  $(\chi^*, \beta)$  severe if  $\chi^*$  is a real primitive character whose conductor  $q^*$  divides some  $q \in \mathcal{Q}$ ,  $L(\beta, \chi^*) = 0$  and  $\beta \geq 1 - 1/(13 \log(3Z^2))$ . Then there is at most one severe pair. If  $(\chi^*, \beta^*)$  is severe and  $q^* \mid q \in \mathcal{Q}$ , then  $\beta_1 = \beta^*$  and  $\chi_1$  is induced by  $\chi^*$  for the modulus  $q$ . If  $q \in \mathcal{Q}$  and no severe pair has  $q^* \mid q$ , then any exceptional  $\beta_1$  of  $q$  satisfies  $x^{\beta_1-1}/\beta_1 \leq 2 \exp(-\log x/(13 \log 3Z^2))$ .*

*Proof.* If  $(\chi_a, \beta_a), (\chi_b, \beta_b)$  are severe with conductors dividing  $q_a, q_b \in \mathcal{Q}$ , put  $q' = \text{lcm}(q_a, q_b) \leq Z^2$ . Both characters induce characters mod  $q'$  with the same zeros in  $\Re s > 0$ , and both  $\beta$  lie in McCurley’s region for  $q'$ ; uniqueness there gives  $\beta_a = \beta_b$  and  $\chi_a = \chi_b$ . The second claim holds because  $\beta^*$  lies in McCurley’s region for  $q \leq Z$ . For the third, an exceptional zero of  $q$  in  $[1 - 1/(13 \log 3Z^2), 1)$  would be severe; also  $\beta_1 \geq 1 - 1/(13 \log 6) > 1/2$ .  $\square$

## 5. A TRANSFER THEOREM

**Theorem 5.1** (Proved modulo Theorem 4.1). *Let  $T \geq 1$ ,  $Q \geq 2$ , and let  $B(n) = \sum_{i \in I} c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$  be a finite real combination with  $\gcd(d_i, Q) = \gcd(b_i, d_i) = 1$  and  $B(n) \leq \mathbf{1}[W(n) > T]$  for all  $n \equiv 1 \pmod{Q}$ . Put  $Z = Q \max d_i$ ,  $\mu = \sum c_i / \varphi(d_i)$ ,  $M_1 = \sum |c_i| / \varphi(d_i)$ , and assume  $\mu > 0$  and the twist condition: for every real primitive character  $\psi$  of conductor  $f > 1$  with  $\gcd(f, Q) = 1$  and  $f \mid d_i$  for some  $i$ ,*

$$|\mu_\psi| \leq \mu/4, \quad \mu_\psi = \sum_{i: f \mid d_i} c_i \psi(b_i) / \varphi(d_i).$$

*Let  $K = 1 + \log(M_1/\mu)$ . There is an absolute  $C_1$  such that if  $\log x \geq C_1 K \max(\log Z, K)$ , then some prime  $p \leq x$ ,  $p \equiv 1 \pmod{Q}$ , has  $W(p) > T$ .*

*Proof.* Let  $q_i = Qd_i$  and let  $a_i$  be the class  $\equiv 1 \pmod{Q}$ ,  $\equiv b_i \pmod{d_i}$ ; it is a unit class. Then  $\sum_{p \leq x, p \equiv 1 \pmod{Q}} \log p B(p) = \sum_i c_i \theta(x; q_i, a_i)$ . Since  $M_1 \geq \mu$  we have  $K \geq 1$ , so taking  $C_1 \geq 12$  gives  $\log x \geq 12 \log Z$ , i.e.  $x \geq Z^{12} \geq q_i^{12}$ , and Theorem 4.1 applies to every term. Put  $E_1 = e^{-c_4 \log x / \log Z} + e^{-c_4 (\log x)^{3/5} / (\log \log x)^{1/5}}$  and  $E_2 = e^{-\log x / (13 \log 3 Z^2)}$ . Since  $\gcd(Q, d_i) = 1$  we have  $\varphi(q_i) = \varphi(Q)\varphi(d_i)$ , and  $q_i \leq Z$ , so

$$\sum_i c_i \theta(x; q_i, a_i) = \frac{x}{\varphi(Q)} \sum_i \frac{c_i \lambda_i (1 + \varepsilon_i)}{\varphi(d_i)}, \quad |\varepsilon_i| \leq C_0 E_1,$$

with  $\lambda_i$  the factor of Theorem 4.1 for  $(q_i, a_i)$ . Apply Lemma 4.2 to  $\mathcal{Q} = \{q_i\}$ ; a severe  $\beta$  satisfies  $\beta \geq 1 - 1/(13 \log 6) > 1/2$ . Write the conductor of a severe  $\chi^*$  as  $q^* = q_Q^* q^{*'} with  $q_Q^*$  supported on the primes of  $Q$  and  $\gcd(q^{*'}, Q) = 1$ , and  $\chi^* = \chi_Q^* \chi^{*'}$ .$

*Case A: a severe pair exists with  $q^* \mid Q$ .* Then  $q^* \mid q_i$  for every  $i$ . By the second claim of Lemma 4.2, every  $\lambda_i$  is formed with the same pair  $(\chi^*, \beta^*)$ , so  $\lambda_i = 1 - \chi^*(a_i) x^{\beta^*-1} / \beta^* = 1 - x^{\beta^*-1} / \beta^* =: \lambda_*$ , because  $a_i \equiv 1 \pmod{q^*}$ . Moreover  $\lambda_* > 0$ : since  $\log x \geq 12 \log Z > 2$  and  $1/2 < \beta^* < 1$ ,  $\log \beta^* \geq 1 - 1/\beta^* > 2(\beta^* - 1) > (\beta^* - 1) \log x$ , i.e.  $x^{\beta^*-1} / \beta^* < 1$ . So  $\sum c_i \theta \geq (x \lambda_* / \varphi(Q))(\mu - C_0 E_1 M_1)$ .

*Case B: otherwise.* If a severe  $\chi^*$  exists, then  $q^*$  divides some  $Qd_j$  with  $\gcd(d_j, Q) = 1$ , so  $q_Q^* \mid Q$ . Hence  $q^* \mid q_i$  iff  $q^{*' \mid d_i}$ , and in Case B we have  $q^{*' > 1$ . For such  $i$ ,  $\chi_1 = \chi^*$  on units mod  $q_i$  (Lemma 4.2), and, as  $a_i \equiv 1 \pmod{q_Q^*}$  and  $a_i \equiv b_i \pmod{q^{*'}}$ ,  $\chi^*(a_i) = \chi_Q^*(1) \chi^{*'}(b_i) = \psi(b_i)$  with  $\psi = \chi^{*'}$ . So  $\lambda_i = 1 - \psi(b_i) x^{\beta^*-1} / \beta^*$ . This  $\psi$  is real primitive (the coprime-conductor factors of a real primitive character are real primitive), of conductor  $q^{*' > 1$  coprime to  $Q$  and dividing some  $d_i$ , so the twist condition applies to it. For all other  $i$  (and for all  $i$  if there is no severe pair),  $|\lambda_i - 1| \leq x^{\beta_1-1} / \beta_1 \leq 2E_2$  by the third claim of Lemma 4.2. Since  $\lambda_i < 2$  and  $x^{\beta^*-1} / \beta^* \leq 1$ , splitting

$\lambda_i = 1 + (\lambda_i - 1)$  gives

$$\begin{aligned} \sum c_i \theta &\geq \frac{x}{\varphi(Q)} (\mu - 2|\mu_\psi| - M_1(2C_0E_1 + 2E_2)) \\ &\geq \frac{x}{\varphi(Q)} (\mu/2 - M_1(2C_0E_1 + 2E_2)), \end{aligned}$$

using  $|\mu_\psi| \leq \mu/4$ . Both brackets are positive once  $E_1, E_2 \leq \mu/(8(C_0 + 1)M_1) = e^{1-K}/(8(C_0 + 1))$ . This holds if  $C_1$  is large:  $\log x/\log Z \geq C_1K$ ,  $(\log x)^{3/5}/(\log \log x)^{1/5} \geq (\log x)^{1/2} \geq C_1^{1/2}K$  and  $\log x/(13 \log 3Z^2) \geq C_1K/52$  (as  $Z \geq 2$ ), so each of the three exponentials is at most  $e^{-cC_1^{1/2}K}$  with  $c > 0$  absolute. Hence  $\sum_p \log p \mathbf{1}[W(p) > T] \geq \sum_p \log p B(p) > 0$ .  $\square$

No lower bound for  $1 - \beta^*$  is used: in Case A the exceptional factor  $\lambda_*$  multiplies main term and error alike. A prime number theorem valid only for  $q \leq \exp(c\sqrt{\log x})$  would force  $\log x \gg (\log Z)^2 \geq T$ ; the Linnik-range uniformity is essential.

## 6. PROOF OF THEOREM 1.1

Let  $J$  be the least even integer  $\geq 22S + 10$  and

$$\begin{aligned} B(n) &= \sum_{P \subseteq \mathcal{U}, |P| \leq J-1} (-1)^{|P|} \prod_{\ell \in P} \mathbf{1}[n \bmod \ell \in F_\ell] = \sum_{j=0}^{J-1} (-1)^j \binom{N(n)}{j}, \\ N(n) &= \#\{\ell \in \mathcal{U} : n \bmod \ell \in F_\ell\}. \end{aligned}$$

*Pointwise bound.* For  $N \geq 1$ ,  $\sum_{j \leq J-1} (-1)^j \binom{N}{j} = (-1)^{J-1} \binom{N-1}{J-1} \leq 0$ ; hence  $B \leq \mathbf{1}[N = 0] \leq \mathbf{1}[W > T]$  on  $n \equiv 1 \pmod{Q}$  by Lemma 3.1. Expanding  $\mathbf{1}[n \bmod \ell \in F_\ell] = \sum_{a \in F_\ell} \mathbf{1}[n \equiv a \pmod{\ell}]$  puts  $B$  in the form of Theorem 5.1, with squarefree moduli  $d_P = \prod_{\ell \in P} \ell \leq T^J$  coprime to  $Q$ .

*Mean and mass.* With  $e_j$  the elementary symmetric functions of the  $g(\ell)$ ,  $\mu = \sum_{j < J} (-1)^j e_j$  and  $M_1 = \sum_{j < J} e_j \leq e^S$ . Let  $N$  be a sum of independent Bernoulli( $g(\ell)$ ) variables; the identity above and  $\binom{N-1}{J-1} \leq \binom{N}{J}$  give  $|\mu - V| \leq e_J \leq (eS/J)^J \leq e^{-2J}$ , where  $V = \prod (1 - g(\ell)) \geq e^{-2S}$  (Lemma 3.2). So  $\mu \geq 0.99V$  and  $\log(M_1/\mu) \leq 3S + 1$ .

*Twist.* If  $\psi$  is real primitive of conductor  $f > 1$ , coprime to  $Q$ , dividing some  $d_P$ , then  $f = \prod_{\ell \in P_f} \ell$  with  $r = |P_f| \geq 1$  and  $\psi = (\cdot/f)$ . With  $h(\ell) = \sum_{a \in F_\ell} (a/\ell)/(\ell - 1)$ ,  $|h| \leq g$ ,

$$\mu_\psi = (-1)^r \prod_{\ell \in P_f} h(\ell) \sum_{P' \subseteq \mathcal{U} \setminus P_f, |P'| \leq J-1-r} (-1)^{|P'|} \prod_{P'} g,$$

which vanishes if  $r \geq J$ . The inner sum is  $V' + \varepsilon'$  with  $V' = V/\prod_{P_f} (1 - g) \leq (16/15)^r V$  and  $|\varepsilon'| \leq e_{J-r} \leq S^{J-r}/(J-r)!$ , so, with  $g \leq 1/16$ ,  $|\mu_\psi| \leq 15^{-r} V + 16^{-r} S^{J-r}/(J-r)!$ . The first term is  $\leq V/15$ . If  $r \leq J/2$ , the second is  $\leq (eS/(J-r))^{J-r} \leq (2eS/J)^{J/2} \leq (e/11)^{J/2} \leq e^{-0.69J} \leq e^{-15S-6}$ , as  $\frac{1}{2} \log(11/e) > 0.69$  and  $J \geq 22S + 10$ . If  $r > J/2$ , it is  $\leq$

$16^{-J/2}e^S \leq e^{-1.38(22S+10)+S} \leq e^{-15S-6}$ . Since  $e^{-15S} \leq e^{-2S} \leq V$ , we get  $|\mu_\psi| \leq (1/15 + e^{-6})V \leq 0.07V \leq \mu/4$ .

*Moduli.*  $\log Z \leq \log Q + J\mathcal{L} \leq 3y$  because  $S = T^{o(1)}$  (Lemma 3.3). Theorem 5.1 gives a prime  $p \equiv 1 \pmod{Q}$  with  $W(p) > T$  and  $\log p \leq C_1(3S+2) \cdot 3y \leq \sqrt{T} \exp((3+\log 2+o(1))\mathcal{L}/\log \mathcal{L})$ . Since  $840 \mid Q$  the prime is hard, and  $p > Q > T$  gives infinitely many distinct primes. Inverting,  $\mathcal{L} \geq 2 \log \log p - O(\log \log p / \log \log \log p)$ . All inputs are effective.  $\square$

## 7. LIMITS OF THE METHOD

**7.1. Complete certificates.** A *complete certificate* for  $W > T$  is a reduced class  $c \pmod{Q}$  all of whose primes have  $W(p) > T$ . The class of one modulo  $L^*(T) = \text{lcm}(24, M \leq T, M \equiv 3 \pmod{4})$  is one, with  $\log L^*(T) = (2/3 + o(1))T$ .

**Proposition 7.1** (Proved). *Every complete certificate has  $\log Q \geq (2/3 - o(1))T$ .*

*Proof.* (i) If a prime  $\ell \equiv 3 \pmod{4}$ ,  $\ell \leq T$ , does not divide  $Q$ , then  $c$  is compatible with the atom  $-4 \in \mathcal{R}(\ell)$  ( $D = 1$ ), and Dirichlet's theorem gives primes with  $W \leq \ell$ ; so  $\prod_{\ell \leq T, \ell \equiv 3(4)} \ell \mid Q$ . (ii) Since  $\mathcal{R}(3) = \{2\}$ ,  $3 \mid Q$  and  $c \equiv 1 \pmod{3}$ . (iii) Let  $\ell \equiv 1 \pmod{4}$ ,  $\ell \leq T/3$ , and suppose a prime  $r \equiv 2 \pmod{3}$  divides  $A = (3\ell + 1)/4$ . The atom  $M = 3\ell$ ,  $D = r$  has class  $-4r \equiv 1 \pmod{3}$ ; if  $\ell \nmid Q$  it is compatible with  $c$ , a contradiction. (iv) The primes  $\ell \equiv 1 \pmod{4}$  for which  $(3\ell + 1)/4$  has no prime factor  $\equiv 2 \pmod{3}$  avoid the classes  $\ell \equiv -3^{-1} \pmod{r}$  for all primes  $r \equiv 2 \pmod{3}$ ,  $r \geq 5$ ; by the prime number theorem in progressions their logarithmic weight up to  $X$  is  $o(X)$ . Hence  $\log Q \geq T/2 + T/6 - o(T)$ .  $\square$

**Remark 7.2** (The previous record; proved modulo [2, Cor. 11]). Put  $L^*(T) = \text{lcm}(24, M \leq T, M \equiv 3 \pmod{4})$ . Every prime  $p \equiv 1 \pmod{L^*(T)}$  has  $W(p) > T$ , by Lemmas 2.1 and 2.2. A prime  $\ell \leq T/3$  divides some such  $M$  (namely  $\ell$  or  $3\ell$ ). A prime  $\ell \in (T/3, T]$  divides one iff  $\ell \equiv 3 \pmod{4}$ . Higher prime powers contribute  $O(\sqrt{T} \log T)$ . Hence  $\log L^*(T) = (2/3 + o(1))T$ . Every prime factor  $\ell$  of  $L^*(T)$  has  $\log \ell \leq \log T = o(\log L^*(T))$ , so Chang's bound gives a prime  $p \equiv 1 \pmod{L^*(T)}$  with  $\log p \leq (12/5 + o(1)) \log L^*(T)$ , i.e.  $W(p) > T \geq (5/8 - \varepsilon) \log p$ . As  $840 \mid L^*(T)$  and  $p > L^*(T) > T$ , this holds for infinitely many hard primes.

Proposition 7.1 bounds  $\log Q$ , not  $p$ : a class  $c \neq 1$  may contain primes below  $Q$ . What it shows is that a complete certificate whose prime is supplied by a least-prime theorem, which certifies  $\log p$  only through an upper bound  $O(\log Q)$ , certifies no more than  $W(p) \gg \log p$ ; with Chang's exponent, no more than  $5/8$  by this route.

**7.2. Prime-local designs.** Call a design *prime-local* if it imposes  $n \equiv 1 \pmod{\ell^{\epsilon_\ell}}$  for  $\ell$  in a set  $\Pi$  and every  $M \leq T$ ,  $M \equiv 3 \pmod{4}$ , has at most one prime factor outside  $\Pi$ . Theorem 1.1 uses such a design.

**Proposition 7.3** (Proved). *In a prime-local design,  $\Pi$  contains every odd prime  $\leq \sqrt{T/3}$  with at most one exception. Hence every prime it produces has  $\log p \geq (1/\sqrt{3} - o(1))\sqrt{T}$ .*

*Proof.* For odd primes  $\ell_1 < \ell_2 \leq \sqrt{T/3}$  outside  $\Pi$ , one of  $\ell_1\ell_2, 3\ell_1\ell_2$  is  $\equiv 3 \pmod{4}$  and  $\leq T$ .  $\square$

### 7.3. What would suffice beyond exponent two.

**Hypothesis 7.4** ( $H_{\min}(\theta)$ ; open for  $\theta < 1/2$ ). For every  $\varepsilon > 0$  and large  $T$  there are  $Q$  with  $840 \mid Q$ ,  $\log Q \leq T^{\theta+\varepsilon}$ , and a minorant  $B$  as in Theorem 5.1 with  $\log \max d_i \leq T^{\theta+\varepsilon}$ ,  $\log(M_1/\mu) \leq T^\varepsilon$ ,  $\mu > 0$ , and the twist condition.

**Theorem 7.5** (Proved modulo Theorem 4.1).  *$H_{\min}(\theta)$  implies  $W(p) > (\log p)^{1/\theta-\varepsilon}$  for infinitely many hard primes, for every  $\varepsilon > 0$ .*

*Proof.* Adjoin to  $Q$  a prime  $\ell_0 \in (R, 2R]$ ,  $R = \max(T, \max d_i)$ ; this keeps all hypotheses and forces  $p > T$ . Theorem 5.1 gives  $\log p \leq T^{\theta+2\varepsilon+o(1)}$ .  $\square$

$H_{\min}$  is a purely combinatorial statement about congruences. For  $\theta < 1/2$  the residual atoms involve several free primes, and plain event-level Bonferroni truncation over the raw atom list fails.

**Proposition 7.6** (Proved). *Fix  $0 < \theta < 1/2$ , put  $y = T^\theta$ , and impose the class of one modulo all prime powers  $\leq T$  of primes  $\leq y$ . Take as events all atoms  $(M, c)$ ,  $c \in \mathcal{R}(M)$ , compatible with this, and let  $N(n)$  count those containing  $n$ . Then for large  $T$  and every even  $J$  with  $(4.2\theta^2 + o(1))(\log T)^2 \leq J \leq y$ , the Bonferroni minorant  $B_J = \sum_{j < J} (-1)^j \binom{N}{j}$  has negative mean in the CRT measure on the class 1.*

*Proof.* The mean is  $P(N=0) - E[(\binom{N-1}{J-1} \mathbf{1}_{N \geq 1})]$ . Let  $r = \lceil \sqrt{2J-1} \rceil \leq \sqrt{2y}+1$ , and take  $r$  primes  $\equiv 1 \pmod{4}$  and  $r$  primes  $\equiv 3 \pmod{4}$  in  $(y, 2y]$ ; there are enough of them. If  $n \equiv -4$  modulo all of them, then for every cross pair  $M = \ell\ell' \leq 4y^2 \leq T$  we have  $M \equiv 3 \pmod{4}$ , and  $D = 1$  gives the atom  $-4 \pmod{M}$ . So  $N \geq r^2 \geq 2J-1$ , on a set of measure  $\geq (2y)^{-2r}$ . Hence

$$E(\binom{N-1}{J-1}) \geq (2y)^{-2r} \binom{2J-2}{J-1} \geq \exp((J-1)\log 4 - \log(2J-1) - 2r(\theta \log T + \log 2)),$$

which exceeds  $1 > P(N=0)$  once  $J > (2\sqrt{2}\theta/\log 4)^2(\log T)^2(1+o(1))$ .  $\square$

These raw hub atoms are implied by single-prime atoms. Numerically, other hub classes survive the deletion of implied atoms, so a hub-adapted (hypergraph) minorant appears necessary; this is not proved.

## 8. THE SLICE PARAMETER

For a prime  $p$  let  $\mathcal{B}_p$  be the set of pairs  $(c, k)$  with  $1 \leq k \leq \lfloor 2p/3 \rfloor$ ,  $1 \leq c \leq \lfloor (2p+k)/(4k) \rfloor$ ,  $(p, ck) = 1$ ; put  $h = 4ck$ ,  $N_{c,k} = p^2 + 4ck^2$ ,  $s = \text{sf}(c)$ , and let  $\chi_s = (\Delta_s/\cdot)$ , where  $\Delta_s < 0$  is the discriminant of  $\mathbb{Q}(\sqrt{-s})$ ; its conductor divides  $4s \mid h$ .  $M_{c,k}(p)$  is the number of divisors of  $N_{c,k}$  in the class  $-p \pmod{h}$ . The slice  $(c, k)$  is *positive* if  $p(a+b) = k(4abc-1)$  has

a solution in positive integers  $a, b$ . This happens iff  $M_{c,k}(p) > 0$ . Both directions rest on the identity, for  $e = ha - p$ ,  $f = hb - p$ ,

$$(ha - p)(hb - p) - N = h(4abck - p(a + b)) - 4ck^2 = h(k(4abc - 1) - p(a + b)),$$

so  $ef = N$  iff  $k(4abc - 1) = p(a + b)$ . If  $e \equiv -p \pmod{h}$  is a positive divisor of  $N$ , then  $f = N/e \equiv p^2(-p)^{-1} \equiv -p \pmod{h}$  (as  $\gcd(p, h) = 1$ ), so  $a = (p + e)/h$ ,  $b = (p + f)/h$  are positive integers with  $ef = N$ . Conversely, if  $k(4abc - 1) = p(a + b)$  then  $pa, pb < 4abck$ , so  $e = ha - p > 0$  and  $f = hb - p > 0$ ; and  $ef = N$ ,  $e \equiv -p \pmod{h}$ .

**Lemma 8.1** (Genus obstruction; proved). *If  $p \equiv 1 \pmod{24}$ ,  $(c, k) \in \mathcal{B}_p$  and  $\chi_s(p) = 1$ , then  $M_{c,k}(p) = 0$ .*

*Proof.* Write  $c = st^2$ . Every prime  $q \mid N$  is odd, prime to  $stk$ , and  $(p(2tk)^{-1})^2 \equiv -s \pmod{q}$ , so  $\chi_s(q) = 1$ ; hence  $\chi_s(e) = 1$  for every divisor  $e$  of  $N$ . But  $\chi_s(-p) = \chi_s(-1)\chi_s(p) = -1$ , and  $\chi_s$  is defined modulo  $h$ .  $\square$

**Lemma 8.2** (A sufficient prime factor; proved). *If a prime  $q$  divides  $N_{c,k}$  and  $q \equiv -p \pmod{h}$ , then  $M_{c,k}(p) \geq 1$ .*

*Proof.*  $q$  itself is a divisor in the class  $-p$ .  $\square$

**Lemma 8.3** (Proved). *Let  $p \equiv 1 \pmod{24}$  and suppose  $(\ell/p) = 1$  for every prime  $5 \leq \ell \leq B$ . Then  $ck_{\min}(p) > B$ . In particular  $ck_{\min}(p) \geq n_p$ , the least quadratic non-residue modulo  $p$ .*

*Proof.* For  $ck \leq B$ ,  $s \leq B$  and  $\chi_s(p) = (-s/p) = (-1/p) \prod_{q \mid s} (q/p) = 1$ , since  $(-1/p) = (2/p) = (3/p) = 1$ . Apply Lemma 8.1.  $\square$

**Proposition 8.4** (Proved). *Let  $24 \mid L$  and let  $a \pmod{L}$  be a reduced class with  $a \equiv 1 \pmod{24}$  such that every sufficiently large prime  $p \equiv a \pmod{L}$  has  $ck_{\min}(p) > T$ . Then every prime  $5 \leq \ell \leq T$  divides  $L$  and  $(a/\ell) = 1$ .*

*Proof.* Suppose  $\ell \nmid L$  or  $(a/\ell) = -1$ . Choose  $c_0 \pmod{4\ell}$  with  $c_0 \equiv a \pmod{\gcd(L, 4\ell)}$ ,  $c_0 \equiv 1 \pmod{4}$ ,  $(c_0/\ell) = -1$ ; for  $n \equiv 1 \pmod{4}$ ,  $\chi_\ell(n) = (n/\ell)$ , so  $\chi_\ell(c_0) = -1$ . Take a prime  $q \equiv -c_0 \pmod{4\ell}$ ,  $q \nmid L$ . Then  $\chi_\ell(q) = \chi_\ell(-1)\chi_\ell(c_0) = 1$ , so  $-4\ell \equiv \rho^2 \pmod{q}$  for some  $\rho \not\equiv 0$ . The conditions  $p \equiv a \pmod{L}$ ,  $p \equiv c_0 \pmod{4\ell}$ ,  $p \equiv \rho \pmod{q}$  are compatible and define a reduced class, which contains infinitely many primes. For these,  $q \mid p^2 + 4\ell$  and  $q \equiv -p \pmod{4\ell}$ , so  $M_{\ell,1}(p) \geq 1$  (Lemma 8.2), and  $(\ell, 1) \in \mathcal{B}_p$  for large  $p$ , whence  $ck_{\min}(p) \leq \ell \leq T$ .  $\square$

**Corollary 8.5** (Proved). *Let  $B(n)$  be a finite combination of residue classes with  $B(p) \leq \mathbf{1}[ck_{\min}(p) > T]$  for all large primes  $p$  in a reduced class  $\equiv 1 \pmod{24}$ . Then every large prime  $p$  in that class with  $B(p) > 0$  has  $n_p > T$ . Together with Lemma 8.3: the depth certifiable by congruences is exactly  $n_p$ .*

*Proof.*  $B$  is periodic modulo some  $L$ , which we take divisible by the modulus of the given class (hence by 24). If  $B(p) > 0$  then  $B$  is positive on the whole

class  $p \bmod L$ , which lies in the given class. So every large prime in the class  $p \bmod L$  has  $ck_{\min} > T$ , and Proposition 8.4 gives  $(p/\ell) = 1$ , i.e.  $(\ell/p) = 1$  by reciprocity ( $p \equiv 1 \pmod{4}$ ), for  $5 \leq \ell \leq T$ ; also  $(2/p) = (3/p) = 1$  as  $p \equiv 1 \pmod{24}$ . Hence  $n_p > T$ .  $\square$

**Theorem 8.6** (Proved modulo [7, Prop. 5.1]).  $ck_{\min}(p) \gg \log p \cdot \log \log \log p$  for infinitely many hard primes  $p$ .

*Proof.* Lau and Wu [7, Prop. 5.1] (following Graham–Ringrose [5]; the numbering is that of the authors’ preprint, and the published numbering was not compared) show the following. They phrase it via  $\chi_p(q) = 1$  with  $\chi_p = (p/\cdot)$ , which for  $p \equiv 1 \pmod{4}$  equals  $(q/p)$  by reciprocity, including  $q = 2$ . The statement: for small fixed  $\delta > 0$  and  $y(x) = \delta \log x \log_3 x$  there are  $x_n \rightarrow \infty$  and primes  $p \in (x_n^{1/2}, x_n \log x_n]$  with  $p \equiv 1 \pmod{4}$  and  $(q/p) = 1$  for all primes  $q \leq y(x_n)$ . Such  $p$  are  $\equiv 1 \pmod{24}$  and, since  $(5/p) = (7/p) = 1$ , lie in a Mordell class. Lemma 8.3 gives  $ck_{\min}(p) > y(x_n) \geq (\delta/2) \log p \log_3 p$ . Effectivity is not claimed.  $\square$

Under GRH, Montgomery’s  $n_p = \Omega(\log p \log \log p)$  would give the corresponding bound, and Ankeny’s  $n_p \ll (\log p)^2$  caps what congruences can certify; the random model predicts  $\max_{p \leq x} n_p \asymp \log x \log \log x$  (statements as quoted in [7]). Finally, the primes of Theorem 1.1 are quadratic residues of every  $\ell \leq y$ , so Lemma 8.3 gives, *jointly*,  $W(p) \geq (\log p)^{2-o(1)}$  and  $ck_{\min}(p) \geq (\log p)^{1-o(1)}$  for infinitely many hard  $p$  (proved modulo Theorem 4.1).

## 9. THE HAAR SIDE

Impose the class of one at all primes  $\leq z$  ( $z \geq 3$ ), i.e.  $n \equiv 1 \pmod{Q_z}$  with  $Q_z$  defined like  $Q$ . For  $M \leq T$  write  $M = mr$  with  $m$  the  $z$ -smooth and  $r$  the  $z$ -rough part. If  $r > 1$ , the surviving atoms of  $M$  are the  $D \mid A_M^2$  with  $m \mid 4D + 1$ , each giving the event  $n \equiv -4D \pmod{r}$  of conditional Haar probability  $1/\varphi(r)$ .

**Lemma 9.1** (Proved). *If  $M \equiv 3 \pmod{4}$ ,  $D \mid A_M^2$  and  $M = mr$  with  $m \mid 4D + 1$ , then  $m \leq r^2 + 1$ .*

*Proof.* By the involution of Lemma 3.3 assume  $D \leq A$ ,  $D = sr'^2$ ,  $A = sr'k$ ,  $k \geq r'$ . As there,  $m \mid r' + k \leq 2k$ , so  $r = (4sr'k - 1)/m \geq 2sr' - 1/(2k)$ , i.e.  $r \geq 2sr'$ , and  $m \leq 4sr'^2 + 1 \leq r^2 + 1$ .  $\square$

**Lemma 9.2** (Proved). *For every  $3 \leq z \leq T$ , the total mass  $S_{\text{tot}} = \sum 1/\varphi(r)$  over all surviving atoms is  $\exp(O(\log T / \log \log T))$ .*

*Proof.* Use  $1/\varphi(r) \ll \log \log T \cdot m/M$  and repeat the proof of Lemma 3.3 with  $\ell$  replaced by  $r$ ; every surviving atom has  $m \mid \gcd(4D + 1, M)$ , hence  $m \mid r' + k$ .  $\square$

(Using [4, Prop. 1.4] on dyadic blocks one gets  $S_{\text{tot}} \ll (\log T)^4 \log \log T$ ; this is not needed below.)

*Proof of Theorem 1.3.* Fix  $\varepsilon > 0$ , put  $y = T^{1/3+\varepsilon}$  and  $z = y$ . Every rough part is  $\ell$ ,  $\ell^2$  or  $\ell_1\ell_2$  (as  $y^3 > T$ ). Use the independent coordinates  $X_\ell = n \bmod \ell^{e_\ell}$ . Single atoms form forbidden sets  $G_\ell$  of measure  $g_\ell$ , with  $1 \notin G_\ell$  (Lemma 2.2) and  $\sum g_\ell \leq S_{\text{tot}}$ .

*Bad primes.*  $B = \{\ell : g_\ell > 1/4\}$  has  $|B| \leq 4S_{\text{tot}} = T^{o(1)}$ . Impose also  $n \equiv 1 \pmod{\ell^{e_\ell}}$  for  $\ell \in B$  (cost  $\leq |B| \log T$ ). Pairs inside  $B$  die by Lemma 2.2; a pair atom  $(\ell b, a)$  with  $b \in B$  becomes the residue  $a \bmod \ell$  at  $\ell$  if  $a \equiv 1 \pmod{\ell}$  (b) and dies otherwise. The enlarged sets satisfy  $g'_\ell \leq 1/4 + |B| \tau_{\text{max}}^2 T / (\ell y (\ell - 1)) \leq 1/2$ , with  $\tau_{\text{max}}^2 = \max_{A \leq T} \tau(A^2) = T^{o(1)}$ , and  $\sum (g'_\ell - g_\ell) \leq |B| \tau_{\text{max}}^2 T / y^2 = T^{1/3-2\varepsilon+o(1)}$ .

*Good pairs.* Under the product measure  $\mu'$  uniform on the allowed residues, a good pair event has  $\mu'(E) \leq 4/((\ell_1 - 1)(\ell_2 - 1))$ , and the per-prime weight is  $w'_\ell \leq 16\tau_{\text{max}}^2 T / (\ell^2 y) \leq T^{-3\varepsilon+o(1)}$ . The asymmetric local lemma [1, Lemma 5.1.1] applies to  $\mu'$ . Here  $\mu'$  is the Haar measure conditioned on a product set, hence a product measure, and a pair event is mutually independent of all events sharing no prime with it. With  $x_E = 2\mu'(E)$  we have  $\sum_{E' \sim E} x_{E'} \leq 2(w'_{\ell_1} + w'_{\ell_2}) = T^{-3\varepsilon+o(1)}$ , so  $\prod_{E' \sim E} (1 - x_{E'}) \geq 1/2$  for large  $T$ . The lemma gives  $\mu'(\text{no good pair event}) \geq \prod (1 - x_E) \geq \exp(-16S_{\text{tot}})$ .

*Collecting.* With  $Q' = Q_y \prod_{b \in B} b^{e_b}$ ,

$$\begin{aligned} \delta^*(T) &\geq \frac{8}{\varphi(Q')} \prod (1 - g'_\ell) e^{-16S_{\text{tot}}} \\ &\geq \exp(-\pi(y) \log T - |B| \log T - 18S_{\text{tot}} - T^{1/3-2\varepsilon+o(1)}) \\ &= \exp(-T^{1/3+\varepsilon+o(1)}). \end{aligned}$$

□

The exponent  $1/3$  reflects the crude bound  $N(M) \leq \tau(A^2)$  for multi-prime atoms at a fixed prime.

**Remark 9.3** (Conditional; the implication is proved modulo [4, Prop. 1.4]). Let  $w_\ell(T, z) = \sum 1/\varphi(r_E)$  over the distinct surviving events  $E$  whose rough part is divisible by  $\ell$ . Suppose  $w_\ell(T, z) \leq \log z / (8 \log T)$  for all primes  $z < \ell \leq T$ , with  $z = (\log T)^C$ , for all large  $T$ . Then  $\log(1/\delta^*(T)) \ll (\log T)^{\max(C+1, 4)+o(1)}$ . Indeed, apply the local lemma on the coordinates  $X_\ell$ ,  $\ell > z$ , to all distinct surviving events, with  $x_E = 2/\varphi(r_E) \leq 1/2$ . An event has at most  $\log T / \log z$  primes, so  $\prod_{E' \sim E} (1 - x_{E'}) \geq \exp(-4 \sum_{\ell | r_E} w_\ell) \geq e^{-1/2}$ , and the lemma gives probability  $\geq \exp(-4S_{\text{tot}})$  of no event. The quarantine costs  $\leq \pi(z) \log T$ , and  $S_{\text{tot}} \ll (\log T)^4 \log \log T$  by [4, Prop. 1.4]. Without [4], Lemma 9.2 gives only  $\delta^*(T) \geq \exp(-T^{o(1)})$  by this argument. We do not know how to prove the per-prime hypothesis.

None of this transfers to primes: the local lemma does not provide a pointwise minorant.

## 10. SUMMARY OF STATUS

- Proved: Lemmas 2.1–3.3, 8.1–8.3, 9.1, 9.2; Propositions 7.1, 7.3, 7.6, 8.4; Corollary 8.5; Theorem 1.3 (with the classical local lemma and divisor bound).
- Proved modulo Thorner–Zaman [14] (with the McCurley statement quoted there): Lemma 4.2, Theorems 5.1, 1.1, 7.5, and the joint  $W/ck_{\min}$  statement.
- Proved modulo Lau–Wu [7, Prop. 5.1]: Theorem 8.6.
- Proved modulo Chang [2, Cor. 11]: the  $5/8$  record (Remark 7.2).
- Conditional on the per-prime hypothesis, and proved modulo [4, Prop. 1.4]: the polylogarithmic Haar bound of Remark 9.3.
- Open:  $H_{\min}(\theta)$  for  $\theta < 1/2$ ; a polylogarithmic Haar bound; any bound  $W(p) > (\log p)^{2+\delta}$  or  $ck_{\min}(p) > (\log p)^{1+\delta}$  infinitely often.

Nothing here bears on the truth of the Erdős–Straus conjecture: a prime with large  $W(p)$  or  $ck_{\min}(p)$  may well have other representations.

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